

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

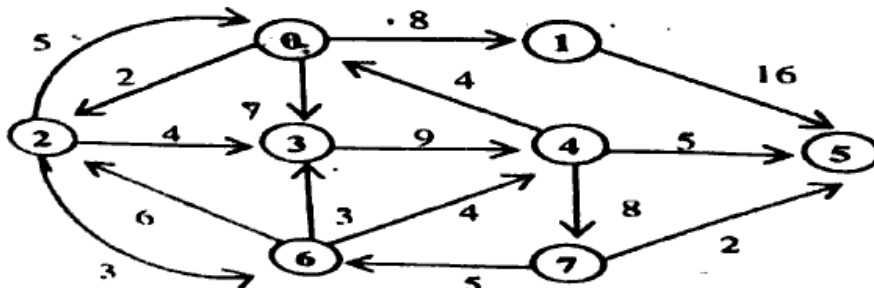
(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

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|---|---|---|
| <p>A) What is asymptotic notation ? Consider a function where time complexity is given by $f(n) = 5n^2 + 6n + 12$. Show the growth rate of all the terms of the function with different inputs and find out the dominant term.</p> <p>B) Define Big O notation, Big Omega and Big Theta notation. Depict the same graphically and explain.</p> <p>C) Order the following functions by growth rate: n, n^2, $n \log n$, 2^n, n^3, $n!$
Also calculate the O notation from the given function.
 i) $f(n) = 7 \log n + 5n + 4$
 ii) $f(n) = n^2 + n^3 + 2^n$
 ii) $f(n) = 23$</p> | <p>3</p> <p>3</p> <p>3</p> | <p>6</p> <p>6</p> <p>6</p> |
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Q.2 Solve Any Two of the following. 12

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|---|---------------------------------|---------------------------------|
| <p>A) What is divide and conquer strategy. Write an algorithm for Binary Search. Also consider an array of 10 elements 10,15,18,20,25,30,49,57,64,72 with searching element as 49 and illustrate the searching process.</p> <p>B) Consider the graph given below. Assuming 0 as source vertex obtain the shortest paths using Dijkstra's algorithm.</p> | <p>3</p> <p>4</p> | <p>6</p> <p>6</p> |
|---|---------------------------------|---------------------------------|



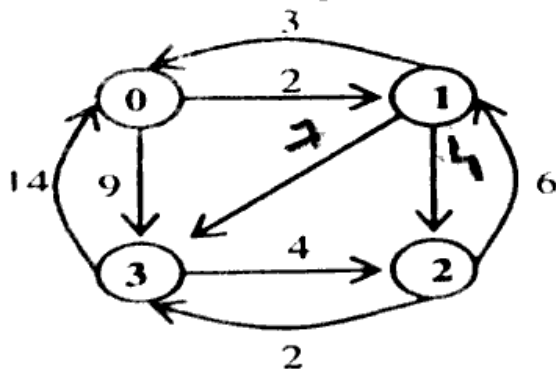
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|--|-----------------|-----------------|
| <p>C) Sort the following numbers using Quick Sort. 48,44,19,59,72,80,42,65,82,8,95,68. Illustrate each step in the sorting process.</p> | <p>4</p> | <p>6</p> |
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Q.3 Solve Any Two of the following.

12

- A) Trace the Floyd's Marshall's algorithm for the given example.

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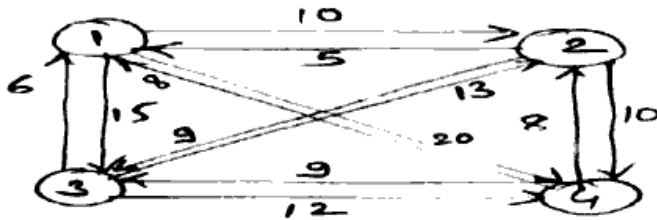
- B) Solve the following knapsack instance using dynamic programming:

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$n=7$, $M=15$, weight $W=(2,3,5,7,1,4,1)$, profit $P=(10,5,15,7,6,18,3)$

- C) Solve the travelling salesman problem using brute force approach for the given example.

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Q.4 Solve Any Two of the following.

12

- A) Compare and contrast DFS and BFS with appropriate example.

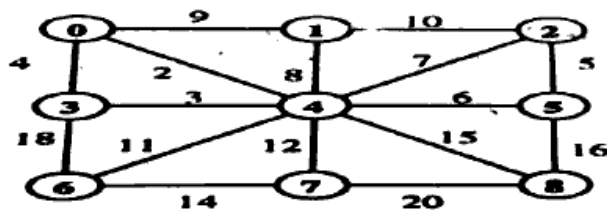
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- B) What is topological sorting. Illustrate it using appropriate example.

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- C) Obtain the minimum cost spanning tree for the following graph using Kruskal's algorithm.

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Q.5 Solve Any Two of the following.

12

- A) What is Backtracking. Consider the box with three different colors with the first object as red color, the second object is of green color, and the third object is of blue color. Generate a state space tree to represent the solution of the above problem.

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- B) Explain in detail about branch and bound technique.

3 6

- C) Draw and explain P, NP, NP Complete and NP hard Problem.

3 6

*** End ***